offered any insight into how it actually works. He gave the world an "owner's manual" for gravity which delineated how to "use" it—instructions that physicists, astronomers, and engineers have exploited successfully to plot the course of rockets to the moon, Mars, and other planets in the solar system; to predict solar and lunar eclipses; to predict the motion of comets, and so on. But he left the inner workings—the contents of the "black box" of gravity—a complete mystery. When you use your CD player or your personal computer, you may find yourself in a similar state of ignorance regarding how it works internally. So long as you know how to operate the equipment neither you nor anyone else needs to know how it accomplishes the tasks you set for it. But if your CD player or personal computer breaks, its repair relies crucially on knowledge of its internal workings. Similarly, Einstein realized that hundreds of years of experimental confirmation notwithstanding, special relativity implied that in some subtle way Newton's theory was "broken" and that its repair required coming to grips with the question of the true and full nature of gravity.

In 1907, while pondering these issues at his desk in the patent office in Bern, Switzerland, Einstein had the central insight that, through fits and starts, would eventually lead him to a radically new theory of gravity—an approach that would not merely fill in the gap in Newton's theory, but, rather, would completely reformulate thinking about gravity and, of utmost importance, would do so in a manner fully consistent with special relativity.

The insight Einstein had is relevant for a question that may have troubled you in Chapter 2. There we emphasized that we were interested in understanding how the world appears to individuals undergoing constant-velocity relative motion. By carefully comparing the observations of such individuals, we found some dramatic implications for the nature of space and time. But what about individuals who are experiencing accelerated motion? The observations of such individuals will be more complicated to analyze than those of constant-velocity observers, whose motion is more serene, but nevertheless we can ask whether there is some way of taming this complexity and bringing accelerated motion squarely into our newfound understanding of space and time.